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Image pickup apparatus, camera and mobile electronic device

Abstract

The present invention discloses an image pickup apparatus, a camera and a mobile electronic device. The image pickup apparatus comprises a housing; a movable component and a fixed component, the fixed component comprising a lens with an optical axis, and the movable component comprising a moving frame and a sensor component; a first connector movably connecting the movable component with the fixed component; a first driving component used for driving the movable component to move in a direction perpendicular to the optical axis; a second connector movably connecting the moving frame with the sensor component; and a second driving component used for driving the sensor component to move in the direction of the optical axis. The image pickup apparatus coincides with the development direction of “miniaturization”, and facilitates optical axis alignment during assembly, thus improving the image pickup quality of the image pickup apparatus.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to the field of optical technology, in particular to an image pickup apparatus, a camera and a mobile electronic device.

BACKGROUND

(2) With the rapid development of science and technology, driving mechanisms for driving lens bodies to move are widely used in various image pickup apparatuses, and the image pickup apparatuses are also widely used in various mobile electronic devices. Here, the driving mechanism of the image pickup apparatus is usually used for driving the lens body to move on an optical axis or even on a plane perpendicular to the optical axis.

(3) The existing driving mechanisms typically comprise coils and magnets. The coil is fixed on a circumference of a lens body frame, so that after the coil is electrified to generate a magnetic field, the lens body moves under an electromagnetic force on the optical axis to realize a focusing

function, and the lens body can even move on the plane perpendicular to the optical axis to realize an anti-shake function.

(4) At present, in case where an optical system has a long total optical length or a glass lens body is adopted to improve the image pickup quality of an image pickup apparatus, the weight of the lens body will be greatly improved. In this case, when the lens body is driven to move by the driving mechanism, the driving mechanism needs to provide a large driving force to be able to drive the heavy lens body to move. Thus, it is necessary to make the driving mechanism and even the image pickup apparatus as large as possible, which is inconsistent with the current development direction of “miniaturization” of image pickup apparatuses.

(5) In addition, in the existing driving mechanism, a focusing drive is arranged in the lens, an anti-shake drive is arranged in a sensor, and it is difficult to align the lens, the focusing drive and the anti-shake drive during assembly, which reduces the optical performance.

(6) Therefore, an image pickup apparatus which coincides with the development direction of “miniaturization”, and also facilitates optical axis alignment during assembly is urgently needed.

SUMMARY

(7) The present invention aims to provide an image pickup apparatus, a camera and a mobile electronic device. The image pickup apparatus coincides with the development direction of “miniaturization”, and also facilitates optical axis alignment during assembly, thus improving the image pickup quality of the image pickup apparatus.

(8) According to the technical scheme of the present invention, an image pickup apparatus comprises a housing with an accommodating space; a movable component accommodated in the housing and a fixed component at least partially accommodated in the housing, the fixed component comprising a lens with an optical axis, and the movable component comprising a moving frame and a sensor component movably connected with the moving frame; a first connector movably connecting the movable component with the fixed component; a first driving component used for driving the movable component to move in a direction perpendicular to the optical axis; a second connector movably connecting the moving frame with the sensor component; and a second driving component used for driving the sensor component to move in the direction of the optical axis.

(9) In some embodiments, the first driving component drives the movable component to move along a first axis and a second axis perpendicular to the optical axis, and the first axis and the second axis are perpendicular to each other.

(10) In some embodiments, the first driving component drives the movable component to rotate relative to the fixed component in the direction of the optical axis.

(11) In some embodiments, the housing comprises an upper housing and a lower housing which cooperate to form the accommodation space, the fixed component comprises a base fixed to the lower housing, the first driving component is a voice coil motor and comprises a first coil and a first magnet, the first coil is arranged on the moving frame, and the first magnet is arranged on the base.

(12) In some embodiments, the second driving component drives the sensor component to rotate around a first axis and a second axis, and the first axis and the second axis are perpendicular to each other and both perpendicular to the optical axis.

(13) In some embodiments, the second driving component comprises four second drive actuators, each second drive actuator comprises a third driver and a fourth driver arranged on the housing, and the fourth driver is used for allowing the third driver to move in a direction close to/away from the fourth driver; and the sensor component has two opposite first side edges and two opposite second side edges, each of the first side edges and each of the second side edges are each provided with one said third driver, the two first side edges are sequentially arranged on the first axis, and the two second side edges are sequentially arranged on the second axis.

(14) In some embodiments, the fixed component comprises a lens holder for accommodating at

least part of the lens, the second driving component is a voice coil motor and comprises a second coil and a second magnet, the second coil is arranged on the sensor component, and the second magnet is arranged on the lens holder.

(15) In some embodiments, the moving frame comprises a moving piece and an anti-shake coil board fixedly connected with the moving piece, and the first connector is a ball and arranged between the moving piece and the base.

(16) In some embodiments, the sensor component comprises an optical filter close to the lens and a sensor circuit board spaced apart from the optical filter, the second connector is a leaf spring, and two ends of the second connector are respectively connected with the anti-shake coil board and the sensor circuit board.

(17) In some embodiments, the image pickup apparatus further comprises a multi-segment accommodating zoom mechanism fixed on the housing.

(18) In some embodiments, the image pickup apparatus further comprises a reflection mechanism fixed on the housing, wherein the lens changes an optical path through the reflection mechanism.

(19) In some embodiments, the image pickup apparatus further comprises an aperture mechanism fixed on the housing and used for adjusting a light inlet amount of the lens.

(20) In some embodiments, the present invention also provides a camera, which comprises the image pickup apparatus.

(21) In some embodiments, the present invention also provides a mobile electronic device, which comprises the camera.

(22) The present invention has the following advantages: to realize the focusing function of the image pickup apparatus, the movable component can be driven through the first driving component to move in the direction of the optical axis, and to realize the anti-shake function of the image pickup apparatus, the sensor component can be driven through the second driving component to move in the direction perpendicular to the optical axis; because for the image pickup apparatus of the present invention, the sensor component with lighter weight instead of a lens body of the lens with heavier weight needs to be driven to move, the first driving component and the second driving component can be miniaturized, so that the image pickup apparatus can be miniaturized too; further, optical axis alignment can be realized easily during assembly, thus improving the image pickup quality of the image pickup apparatus.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a structural diagram of an image pickup apparatus provided by some embodiments of the present invention;

(2) FIG. 2 is an exploded view of an image pickup apparatus provided by some embodiments of the present invention;

(3) FIG. 3 is an exploded view of some parts of an image pickup apparatus provided by some embodiments of the present invention;

(4) FIG. 4 is an exploded view of some other parts of an image pickup apparatus provided by some embodiments of the present invention;

(5) FIG. 5 is a front view of an image pickup apparatus provided by some embodiments of the present invention;

(6) FIG. 6 is a structural diagram of an image pickup apparatus from another perspective provided by some embodiments of the present invention;

(7) FIG. 7 is a sectional view taken along A-A in FIG. 6;

(8) FIG. 8 is a partial enlarged view of part B in FIG. 7;

(9) FIG. 9 is a structural diagram of some structures of an image pickup apparatus provided by

some embodiments of the present invention;

(10) FIG. **10** is a diagram showing a first driving component, driving a movable component to move along a first axis, of an image pickup apparatus provided by some embodiments of the present invention;

(11) FIG. **11** is a diagram showing a first driving component, driving a movable component to move along a second axis, of an image pickup apparatus provided by some embodiments of the present invention;

(12) FIG. **12** is a diagram showing a first driving component, driving a movable component to rotate counterclockwise around a first axis, of an image pickup apparatus provided by some embodiments of the present invention;

(13) FIG. **13** is a diagram showing a first driving component, driving a movable component to rotate clockwise around a first axis, of an image pickup apparatus provided by some embodiments of the present invention;

(14) FIG. **14** is a diagram showing a second driving component of an image pickup apparatus provided by some embodiments of the present invention driving a sensor component to move in the direction of an optical axis;

(15) FIG. **15** is a diagram showing a second driving component, driving a sensor component to rotate around a first axis, of an image pickup apparatus provided by some embodiments of the present invention;

(16) FIG. **16** is a diagram showing a second driving component, driving a sensor component to rotate around a second axis, of an image pickup apparatus provided by some embodiments of the present invention;

(17) FIG. **17** is a structural diagram of a multi-segment accommodating zoom mechanism and a lens of an image pickup apparatus provided by some embodiments of the present invention;

(18) FIG. **18** is a structural diagram of a reflection mechanism and a lens of an image pickup apparatus provided by some embodiments of the present invention;

(19) FIG. **19** is a structural diagram of an aperture mechanism and a lens of an image pickup apparatus provided by some embodiments of the present invention;

(20) FIG. **20** is a structural diagram of a flexible circuit board provided by some embodiments of the present invention; and

(21) FIG. **21** is a structural diagram of a mobile electronic device according to some embodiments of the present invention.

DESCRIPTION OF REFERENCE NUMERALS

(22) housing **110**, upper housing **111**, lower housing **112**; movable component **120**, movable frame **121**, first side **101**, second side **102**, third side **103**, fourth side **104**, sensor component **122**, first side edge **123**, second side edge **124**, moving piece **125**, anti-shake coil board **126**, optical filter **127**, sensing circuit board **128**, first anti-shake adhesive **129**; fixed component **130**, lens **131**, base **132**, lens holder **133**, second anti-shake adhesive **134**; first connector **140**, first support plate **141**, second support plate **142**; first driving component **150**, first drive actuator **151**, first driver **152**, second driver **153**, first magnetic yoke **154**; second connector **160**; second driving component **170**, second drive actuator **171**, third driver **172**, fourth driver **173**, second magnetic yoke **174**; multi-segment accommodating zoom mechanism **180**; reflection mechanism **190**; aperture mechanism **200**; flexible circuit board **210**; position detection member **220**; mobile electronic device **230**; camera **240**.

DETAILED DESCRIPTION OF EMBODIMENTS

(23) The present invention will be further described below with reference to the accompanying drawings and embodiments.

(24) Referring to FIGS. **1-9**, in some embodiments, the present invention provides an image pickup apparatus, comprising: a housing **110** with an accommodating space; a movable component **120** accommodated in the housing **110** and a fixed component **130** at least partially accommodated in

the housing **110**, the fixed component **130** comprising a lens **131** with an optical axis, and the movable component **120** comprising a moving frame **121** and a sensor component **122** movably connected with the moving frame **121**; a first connector **140** movably connecting the movable component **120** with the fixed component **130**; a first driving component **150** used for driving the movable component **120** to move in a direction perpendicular to the optical axis; a second connector **160** movably connecting the moving frame **121** with the sensor component **122**; and a second driving component **170** used for driving the sensor component **122** to move in the direction of the optical axis.

(25) Specifically, the fixed component **130** being at least partially accommodated in the housing **110** means that the whole fixed component **130** can be accommodated in the housing **110**, or part of the fixed component **130** can be accommodated in the housing **110**. It should be noted that the specific arrangement of the fixed component **130** is not limited by the present invention, as long as an optical image signal transmitted through the lens **131** of the fixed component **130** can be irradiated to an imaging surface of the sensor component **122**, so that the sensor component **122** can convert the optical image signal into an electrical signal.

(26) The movable component **120** is accommodated in the housing **110**, so that the housing **110** comprehensively protect the movable component **120** with the sensor component **122**, thus preventing the sensor component **122** from being damaged due to collision with other parts.

(27) Because the first connector **140** movably connects the movable component **120** with the fixed component **130**, the first driving component **150** can drive the movable component **120** to move relative to the fixed component **130** in a direction perpendicular to the optical axis, so that the image pickup apparatus can realize the anti-shake function when the first driving component **150** drives the movable component **120** with the sensor component **122** to move in the direction perpendicular to the optical axis. It should be noted that how exactly the movable component **120** and the fixed component **130** are movably connected by the first connector **140** is not limited by the present invention.

(28) Because the second connector **160** movably connects the moving frame **121** with the sensor component **122**, when the second driving component **170** drives the sensor component **122** to move in the direction of the optical axis, the sensor component **122** can move in the direction of the optical axis relative to the moving frame **121**, so that the sensor component **122** can move in the direction of the optical axis relative to the fixed component **130** with the lens **131**, and the image pickup apparatus can realize the focusing function when the second driving component **170** drives the sensor component **122** to move in the direction of the optical axis. It should be noted that how exactly the moving frame **121** and the sensor component **122** are movably connected by the second connector **160** is not limited by the present invention.

(29) In this way, when the image pickup apparatus provided by the present invention is used, the anti-shake function of the image pickup apparatus can be realized by making the first driving component **150** drive the movable component **120** to move in the direction perpendicular to the optical axis, so that the sensor component **122** of the movable component **120** can move in the direction perpendicular to the optical axis; and the focusing function of the image pickup apparatus can be realized by making the second driving component **170** drive the sensor component **122** to move in the direction of the optical axis.

(30) Because for the image pickup apparatus of the present invention, the light movable component **120** with the sensor component **122** is driven to move and the sensor component **122** is directly driven to move, instead of driving a lens body in the heavy lens **131** to move, the first driving component **150** and the second driving component **170** can make the image pickup apparatus have anti-shake and focusing functions without providing a large driving force, so that the first driving component **150** and the second driving component **170** can be miniaturized, and then the image pickup apparatus can be miniaturized too. Moreover, during the driving process of the first driving component **150** and the second driving component **170**, because the first driving component **150**

and the second driving component **170** do not drive the lens body in the heavy lens **131** to move, the lens body in the lens **131** can be prevented from shaking during focusing of the image pickup apparatus, thereby improving the image pickup quality of the image pickup apparatus.

(31) In addition, since in the present invention, the anti-shake and focusing functions are integrated on the sensor component **122**, that is, the anti-shake and focusing functions are realized by driving the sensor component **122** to move, the difficulty of aligning the lens **131**, a focusing drive and an anti-shake drive can be reduced when assembling the image pickup apparatus.

(32) Still referring to FIGS. **1-9**, in some embodiments, the first driving component **150** drives the movable component **120** to move along a first axis and a second axis perpendicular to the optical axis, and the first axis and the second axis are perpendicular to each other.

(33) Specifically, the first driving component **150** driving the movable component **120** to move along the first axis and the second axis perpendicular to the optical axis means that the first driving component **150** can not only drive the movable component **120** to move along the first axis perpendicular to the optical axis so that the movable component **120** with the sensor component **122** moves along the first axis relative to the fixed component **130** with the lens **131**, but also drive the movable component **120** to move along the second axis perpendicular to the optical axis, so that the movable component **120** with the sensor component **122** moves along the second axis relative to the fixed component **130** with the lens **131**. In this way, when the image pickup apparatus is used, the first driving component **150** can be used for driving the movable component **120** with the sensor component **122** to move along the first axis and the second axis, so as to realize the anti-shake function of the image pickup apparatus.

(34) In addition, while the first driving component **150** drives the movable component **120** to move along the first axis, the first driving component **150** can also drive the movable component **120** to move along the second axis, so that the movable component **120** can move in any direction perpendicular to the optical axis under the driving of the first driving component **150**, which improves the anti-shake effect of the image pickup apparatus.

(35) Still referring to FIGS. **1-9**, in some embodiments, the first driving component **150** drives the movable component **120** to rotate relative to the fixed component **130** in the direction of the optical axis.

(36) Specifically, the first driving component **150** driving the movable component **120** to rotate relative to the fixed component **130** around the optical axis means that the first driving component **150** can drive the movable component **120** to rotate on a plane perpendicular to the optical axis. In this way, when the image pickup apparatus is used, the first driving component **150** can be used for making the movable component **120** with the sensor component **122** rotate relative to the fixed component **130** around the optical axis, thereby further improving the anti-shake effect of the image pickup apparatus.

(37) Still referring to FIGS. **1-9**, in some embodiments, the first driving component **150** comprises four first drive actuators **151**, each first drive actuator **151** comprises a first driver **152** and a second driver **153** arranged on the housing **110**, and the second driver **153** is used for moving the first driver **152** in a direction perpendicular to the optical axis but not intersecting with the optical axis; and the moving frame **121** is in a square plate shape, and has a first side **101**, a second side **102**, a third side **103** and a fourth side **104** connected end to end, the first side **101** is provided with one first driver **152** at a position near the second side **102**, the second side **102** is provided with one first driver **152** at a position near the third side **103**, the third side **103** is provided with one first driver **152** at a position near the fourth side **104**, and the fourth side **104** is provided with one first driver **152** at a position near the first side **101**.

(38) In this way, still referring to FIGS. **1-9** and also referring to FIG. **10**, the first driver **152** arranged on the first side **101** and the first driver **152** arranged on the third side **103** are driven to move in a direction (that is, the direction indicated by the arrows drawn at the two first drivers **152** in FIG. **10**) parallel to the first axis (that is, the X axis in FIG. **10**) by two second drivers **153**

respectively corresponding to the first driver **152** arranged on the first side **101** and the first driver **152** arranged on the third side **103** (that is, the two first drivers **152** in FIG. **10**), that is, the moving frame **121** moves along the first axis, and the sensor component **122** is driven by the movement of the moving frame **121** to move along the first axis.

(39) Still referring to FIGS. **1-9** and also referring to FIG. **11**, the first driver **152** arranged on the second side **102** and the first driver **152** arranged on the fourth side **104** are driven to move in a direction (that is, the direction indicated by the arrows drawn at the two first drivers **152** in FIG. **11**) parallel to the second axis (that is, the Y axis in FIG. **11**) by two second drivers **153** respectively corresponding to the first driver **152** arranged on the second side **102** and the first driver **152** arranged on the fourth side **104** (that is, the two first drivers **152** in FIG. **11**), that is, the moving frame **121** moves along the second axis, and the sensor component **122** is driven by the movement of the moving frame **121** to move along the second axis.

(40) In addition, the four second drivers **153** can simultaneously drive the four first drivers **152** respectively arranged on the four sides to move in a direction close to or away from the optical axis, so that the moving frame **121** rotates, and the sensor component **122** is driven by the rotation of the moving frame **121** to rotate.

(41) Specifically, still referring to FIGS. **1-9** and also referring to FIG. **12**, the four second drivers **153** simultaneously drive the four first drivers **152** respectively arranged on the four sides to move in the direction close to the optical axis (that is, the direction indicated by the arrows drawn at the four first drivers **152** in FIG. **12**), so that the moving frame **121** rotates counterclockwise, and the sensor component **122** is driven by the counterclockwise rotation of the moving frame **121** to rotate counterclockwise.

(42) Still referring to FIGS. **1-9** and also referring to FIG. **13**, the four second drivers **153** simultaneously drive the four first drivers **152** respectively arranged on the four sides to move in the direction away from the optical axis (that is, the direction indicated by the arrows drawn at the four first drivers **152** in FIG. **13**), so that the moving frame **121** rotates clockwise, and the sensor component **122** is driven by the clockwise rotation of the moving frame **121** to rotate clockwise.

(43) Still referring to FIGS. **1-9**, in some embodiments, the housing **110** comprises an upper housing **111** and a lower housing **112** which cooperate to form the accommodation space, the fixed component **130** comprises a base **132** fixed to the lower housing **112**, the first driving component **150** is a voice coil motor and comprises a first coil and a first magnet, the first coil is arranged on the moving frame **121**, and the first magnet is arranged on the base **132**.

(44) In some embodiments, the first magnet comprises two first magnetic blocks arranged in sequence in a driving direction of the voice coil motor (that is, the direction perpendicular to the optical axis), the N pole and S pole of any one of the two first magnetic blocks are arranged in sequence in the direction of the optical axis, and the two first magnetic blocks have opposite magnetisms.

(45) In some embodiments, the base **132** is provided with positioning grooves in one-to-one correspondence to the first magnets, and the positioning grooves are used for accommodating the first magnets so that the first magnets are arranged on the base **132**.

(46) In some embodiments, the first driving component **150** comprises a plurality of first drive actuators **151**. The first drive actuator **151** comprises a first driver **152** and a second driver **153**. When the first driving component **150** is a voice coil motor, that is, when the first drive actuator **151** is a voice coil motor, the first driver **152** is one of the first coil and the first magnet, and the second driver **153** is the other of the first coil and the first magnet. Further, the first drive actuator **151** further comprises a first magnetic yoke **154** which is arranged opposite to the first magnet. When a coil of the voice coil motor is electrified, because the position of the magnet is fixed, the magnetic field generated by the magnet and the magnetic field generated when the coil is electrified can be used for making the coil move in a direction away from/close to the magnet (that is, the driving direction of the voice coil motor); and when the coil of the voice coil motor is not

electrified, because the position of the magnet is fixed, the magnetic field generated by the magnet and the magnetic field generated by the magnetic yoke can be used for making the magnetic yoke move in the direction close to the magnet, so as to realize position recovery.

(47) Still referring to FIGS. 1-9, in some embodiments, the second driving component 170 drives the sensor component 122 to rotate around the first axis and the second axis. Specifically, the second driving component 170 driving the sensor component 122 to rotate around the first axis and the second axis means that the second driving component 170 can drive the sensor component 122 to rotate around the first axis and the second driving component 170 can also drive the sensor component 122 to rotate around the second axis. In this way, when the image pickup apparatus is used, the second driving component 170 can be used for driving the sensor component 122 to rotate around the first axis and the second axis, so as to realize the anti-shake function of the image pickup apparatus.

(48) Still referring to FIGS. 1-9, in some embodiments, the second driving component 170 comprises four second drive actuators 171, each second drive actuator 171 comprises a third driver 172 and a fourth driver 173 arranged on the housing 110, and the fourth driver 173 is used for allowing the third driver 172 to move in a direction close to/away from the fourth driver 173; and the sensor component 122 has two opposite first side edges 123 and two opposite second side edges 124, each of the first side edges 123 and each of the second side edges 124 are each provided with one said third driver 172, the two first side edges 123 are sequentially arranged on the first axis, and the two second side edges 124 are sequentially arranged on the second axis.

(49) In this way, still referring to FIGS. 1-9 and also referring to FIG. 14, when the sensor component 122 needs to be driven by the second driving component 170 to move in the direction of the optical axis, the four third drivers 172 can be driven by the four fourth drivers 173 to move in the direction of the optical axis (that is, the direction indicated by the arrows drawn at the four third drivers 172 in FIG. 14). In this way, the focusing function of the image pickup apparatus is realized.

(50) Still referring to FIGS. 1-9 and also referring to FIG. 15, when the sensor component 122 needs to be driven to rotate around the first axis (that is, the X axis in FIG. 15) by the second driving component 170, the third drivers 172 respectively arranged on the two second side edges 124 (that is, the two third drivers 172 in FIG. 15) can be driven to move in opposite directions (that is, the directions indicated by the arrows drawn at the two third drivers 172 respectively in FIG. 15) by the two fourth drivers 173 respectively corresponding to the third drivers 172 respectively arranged on the two second side edges 124.

(51) Still referring to FIGS. 1-9 and also referring to FIG. 16, when the sensor component 122 needs to be driven by the second driving component 170 to rotate around the second axis (that is, the Y axis in FIG. 16), the third drivers 172 respectively arranged on the two first side edges 124 (that is, the two third drivers 172 in FIG. 16) can be driven to move in opposite directions (that is, the directions indicated by the arrows drawn at the two third drivers 172 respectively in FIG. 16) by the two fourth drivers 173 respectively corresponding to the third drivers 172 respectively arranged on the two first side edges 124. In this way, the anti-shake effect of the image pickup apparatus can be further improved.

(52) Still referring to FIGS. 1-9, in some embodiments, the third driver 172 and the fourth driver 173 are sequentially arranged in a direction parallel to the optical axis. In this way, the direction of a driving force provided by the second drive actuator 171 can be made parallel to the optical axis, thereby facilitating the anti-shake function of the image pickup apparatus.

(53) Still referring to FIGS. 1-9, in some embodiments, the fixed component 130 comprises a lens holder 133 for accommodating at least part of the lens 131, the second driving component 170 is a voice coil motor and comprises a second coil and a second magnet, the second coil is arranged on the sensor component 122, and the second magnet is arranged on the lens holder 133.

(54) In some embodiments, the second magnet comprises a second magnetic block, and the N pole

and the S pole of the second magnetic block are sequentially arranged in the direction parallel to the optical axis.

(55) In some embodiments, the lens holder **133** is provided with limiting grooves in one-to-one correspondence to the second magnets, and the limiting grooves are used for accommodating the second magnets so that the second magnets are arranged on the lens holder **133**.

(56) In some embodiments, the second driving component **170** comprises a plurality of second drive actuators **171**, and the second drive actuator **171** comprises a third driver **172** and a fourth driver **173**. When the second driving component **170** is a voice coil motor, that is, when the second drive actuator **171** is a voice coil motor, the third driver **172** is one of the second coil and the second magnet, and the fourth driver **173** is the other of the second coil and the second magnet. Further, the second drive actuator **171** further comprises a second magnetic yoke **174** arranged on the second coil.

(57) Still referring to FIGS. **1-9**, in some embodiments, the first driving component **150** and the second driving component **170** are staggered in the direction parallel to the optical axis. In this way, the first driving component **150** and the second driving component **170** may overlap in the direction perpendicular to the optical axis, thereby increasing the utilization rate of the space in the housing **110**.

(58) Still referring to FIGS. **1-9**, in some embodiments, the moving frame **121** comprises a moving piece **125** and an anti-shake coil board **126** fixedly connected with the moving piece **125**. The first connector **140** is a ball and arranged between the moving piece **125** and the base **132**. In this way, the moving piece **125** can slide in any direction perpendicular to the optical axis relative to the housing **110**, and the moving piece **125** can rotate around the optical axis.

(59) In some embodiments, the first magnetic yoke **154** is arranged on the moving piece **125**.

(60) In some embodiments, the image pickup apparatus further comprises a first support plate **141** fixed on the moving piece **125** and a second support plate **142** fixed on the base **132**, and the first support plate **141** and the second support plate **142** are opposite on two sides of the ball and are slidably connected with the ball.

(61) Still referring to FIGS. **1-9**, in some embodiments, the sensor component **122** comprises an optical filter **127** close to the lens **131** and a sensor circuit board **128** spaced apart from the optical filter **127**, the second connector **160** is a leaf spring, and two ends of the second connector are respectively connected with the anti-shake coil board **126** and the sensor circuit board **128**.

(62) In this way, the sensor circuit board **128** can move along the optical axis and rotate around the first axis and the second axis. It should be noted that in other embodiments, the second connector **160** can also be a spring and other elements that can be elastically deformed, which is not limited by the present invention. It should also be noted that the specific number of the second connectors **160** is not limited in the present invention.

(63) In some embodiments, the optical filter **127** can be fixed on the sensor circuit board **128** by gluing, screwing and other fixing methods.

(64) In some embodiments, the third driver **172** is arranged on the sensor circuit board **128**.

(65) In some embodiments, the moving frame **121** is coated with a first anti-shake adhesive **129**. The first anti-shake glue **129** can suppress the shaking of the image pickup apparatus under abrupt electrification, thus achieving the anti-shake effect, and further improving the anti-shake effect of the image pickup apparatus.

(66) In some embodiments, the base **132** is coated with a second anti-shake adhesive **134**. The second anti-shake adhesive **134** can also suppress the shaking of the image pickup apparatus under abrupt electrification, thus achieving the anti-shake effect, and further improving the anti-shake effect of the image pickup apparatus.

(67) Still referring to FIG. **2** and also referring to FIG. **17**, in some embodiments, the image pickup apparatus further comprises a multi-segment accommodating zoom mechanism **180** fixed on the housing **110**. Specifically, the multi-segment accommodating zoom mechanism **180** is used for

adjusting a focal length of the lens **131**. The multi-segment accommodating zoom mechanism **180** is provided with a plurality of lens barrels arranged in a nested manner, and the plurality of lens barrels can be extended or shortened in the direction of the optical axis, thereby increasing or decreasing the distance between the lens **131** and the sensor component **122** to realize zooming. (68) Referring to FIG. **18**, in some embodiments, the image pickup apparatus further comprises a reflection mechanism **190** fixed on the housing **110**, wherein the lens **131** changes an optical path through the reflection mechanism **190**. Specifically, the reflecting mechanism **190** can be a mechanism with a reflecting mirror, and the reflecting mirror can reflect light to change the optical path.

(69) Referring to FIG. **19**, in some embodiments, the image pickup apparatus further comprises an aperture mechanism **200** fixed on the housing **110** and used for adjusting a light inlet amount of the lens **131**. In this way, the light inlet amount of the lens **131** can be adjusted through the aperture mechanism **200**, thereby changing the brightness of images captured by the image pickup apparatus.

(70) In addition, because in the related art, a lens body of a lens is driven by a driving mechanism to move to realize the anti-shake or zoom function, an image pickup apparatus provided by the related art will cause a change in the position of the center of gravity of the lens during the anti-shake or zoom operation, and it is not suitable for the existing image pickup apparatus to be provided with an aperture mechanism on the lens which can adjust the light inlet amount of the lens. However, in the image pickup apparatus provided by the present invention, the position of the center of gravity of the lens **131** will not change during the anti-shake or zoom operation of the image pickup apparatus, so the aperture mechanism **200** can be installed on the lens **131**.

(71) In some embodiments, the present invention provides a camera, comprising an image pickup apparatus. In fact, the image pickup apparatus comprised by the camera provided by the present invention is the same as the image pickup apparatus in the previous embodiment, so the camera provided by the present invention has the same beneficial effects as the image pickup apparatus in the previous embodiment, which will not be repeated here.

(72) Still referring to FIGS. **1-9** and also referring to FIG. **20**, in some embodiments, the image pickup apparatus further comprises a flexible circuit board **210**. The flexible circuit board **210** has a first conductive end **211**, a second conductive end **212**, and a body **213** connected with the first conductive end **211** and the second conductive end **212**. The first conductive end **211** is located in the housing **110**, the second conductive end **212** is located outside the housing **110**, and the body **213** surrounds the optical axis and is arranged along a peripheral edge of the sensor component **122** around the optical axis. In this way, power can be supplied to any place in the housing **110** through the first conductive end **211** of the flexible circuit board **210**, and the flexible circuit board **210** will not interfere with the sensor component **122** moving under the driving of the first driving component **150** and the second driving component **170**. Moreover, the flexible circuit board **210** will not affect the irradiation of the light passing through the lens **131** on an imaging surface of the sensor component **122**.

(73) Still referring to FIG. **1-9**, in some embodiments, the image pickup apparatus further comprises a position detection member **220** arranged on the sensor component **122**. The position detection member **220** is used for detecting the displacement and/or the direction and angle of rotation of the sensor component **122** relative to the lens **131**.

(74) In this way, when the sensor component **122** is driven to move and/or rotate relative to the lens **131** by the first driving component **150** and the second driving component **170**, the position detection member **220** can be used for detecting the displacement and/or the direction and angle of rotation of the sensor component **122** relative to the lens **131**, so as to determine whether the sensor component **122** is driven to a target position by the first driving component **150** and the second driving component **170**. In one example, there are a plurality of position detection members **220**, and each first driver **152** and each third driver **172** are each provided with one position detection

member 220.

(75) Referring to FIG. 21, in some embodiments, the present invention provides a mobile electronic device 230 comprising a camera 240. In fact, the camera 240 comprised by the mobile electronic device 230 provided by the present invention is the same as the camera in the above embodiment, and the camera in the above embodiment has the image pickup apparatus in the aforementioned embodiment, so the mobile electronic device 230 provided by the present invention has the same beneficial effects as the image pickup apparatus in the aforementioned embodiment, which are not repeated here.

(76) The specific type of the mobile electronic device 230 is not limited by the present invention. For example, the mobile electronic device 230 may be a mobile phone, a tablet, etc.

(77) The above are only embodiments of the present invention, and it should be pointed out here that for those of ordinary skill in the art, improvements can be made without departing from the inventive concept of the present invention, and all these improvements fall within the protection scope of the present invention.

Claims

1. An image pickup apparatus, comprising: a housing with an accommodating space; a movable component accommodated in the housing and a fixed component at least partially accommodated in the housing, the fixed component comprising a lens with an optical axis, and the movable component comprising a moving frame and a sensor component movably connected with the moving frame; a first connector movably connecting the movable component with the fixed component; a first driving component used for driving the movable component to move in a direction perpendicular to the optical axis and driving the movable component to rotate relative to the fixed component in a direction of the optical axis; a second connector movably connecting the moving frame with the sensor component; and a second driving component used for driving the sensor component to move in the direction of the optical axis.
2. The image pickup apparatus according to claim 1, wherein the first driving component drives the movable component to move along a first axis and a second axis perpendicular to the optical axis, and the first axis and the second axis are perpendicular to each other.
3. The image pickup apparatus according to claim 1, wherein the housing comprises an upper housing and a lower housing which cooperate to form the accommodation space, the fixed component comprises a base fixed to the lower housing, the first driving component is a voice coil motor and comprises a first coil and a first magnet, the first coil is arranged on the moving frame, and the first magnet is arranged on the base.
4. The image pickup apparatus according to claim 3, wherein the moving frame comprises a moving piece and an anti-shake coil board fixedly connected with the moving piece, and the first connector is a ball and arranged between the moving piece and the base.
5. The image pickup apparatus according to claim 4, wherein the sensor component comprises an optical filter close to the lens and a sensor circuit board spaced apart from the optical filter, the second connector is a leaf spring, and two ends of the second connector are respectively connected with the anti-shake coil board and the sensor circuit board.
6. The image pickup apparatus according to claim 1, wherein the second driving component drives the sensor component to rotate around a first axis and a second axis, and the first axis and the second axis are perpendicular to each other and both perpendicular to the optical axis.
7. The image pickup apparatus according to claim 6, wherein the second driving component comprises four second drive actuators, each of the second drive actuators comprises a third driver and a fourth driver arranged on the housing, and the fourth driver is used for allowing the third driver to move in a direction close to/away from the fourth driver; and the sensor component has two opposite first side edges and two opposite second side edges, each of the first side edges and

each of the second side edges are each provided with one said third driver, the two first side edges are sequentially arranged on the first axis, and the two second side edges are sequentially arranged on the second axis.

8. The image pickup apparatus according to claim 1, wherein the fixed component comprises a lens holder for accommodating at least part of the lens, the second driving component is a voice coil motor and comprises a second coil and a second magnet, the second coil is arranged on the sensor component, and the second magnet is arranged on the lens holder.

9. The image pickup apparatus according to claim 1, further comprising a multi-segment accommodating zoom mechanism fixed on the housing.

10. The image pickup apparatus according to claim 1, further comprising a reflection mechanism fixed on the housing, wherein the lens changes an optical path through the reflection mechanism.

11. The image pickup apparatus according to claim 1, further comprising an aperture mechanism fixed on the housing and used for adjusting a light inlet amount of the lens.

12. A camera, comprising the image pickup apparatus according to claim 1.

13. A mobile electronic device, comprising the camera according to claim 12.
